

Amit Chandran

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Multidisciplinary systems engineer with experience developing C++ firmware, low-latency telemetry pipelines, and fully integrated electro-mechanical systems. Proven ability to deliver high-precision software and hardware solutions under strict performance and regulatory constraints

Education

UCL - MEng Mechanical Engineering

Expected June 2028

- Relevant Modules: Thermodynamics, Dynamics, Mathematical Modelling, Robotics, Control Systems
- Current Average: 78% - First Class

Experience

Systems Engineering Intern – Leonardo UK

Summer 2025 and 2026

- Operated as a systems engineer, coordinating 10 other interns to seamlessly integrate antenna, firmware, hardware and software components into a continuous RF to web-server pipeline.
- Developed a compression algorithm reducing FFT data bandwidth requirements by over **2500×**, minimising network utilisation whilst preserving critical information
- Optimised signal max-hold algorithm using MATLAB by reducing complexity from $O(N^2)$ to $O(N)$ achieving a **99.3%** simulation time reduction
- Streamlined obsolete manual signal analysis by implementing a fully automated MATLAB pipeline, reducing processing time from several days to **under 5 minutes**
- Launched a student led initiative at the company, promoting more cross-disciplinary collaboration and knowledge sharing between functions. Now being implemented into the scheme for future years

Projects / Extracurricular

Technical Lead and Drone Controls Engineer – UCL Rover Team

January 2025 – present

- Programmed an edge-computing vision pipeline in Python and OpenCV on a Raspberry Pi to track ArUco markers in real-time, with an average pose error of **<1cm**
- Established a robust UDP telemetry link to stream positional data seamlessly from the edge device to a high-level MATLAB/Simulink control model
- Built a UART-based MAVLink telemetry architecture connecting Simulink models directly to an H743 flight controller for real-time flight control

Smart Pet Feeder – BLE IoT System

October - March 2026

- Developed C++ firmware on an ESP32-S3, interfacing a NEMA17 stepper motor and HX711 strain-gauge amplifier for real-time mass measurement and portion control
- Implemented an event-driven BLE GATT server (Nordic UART Service) that dynamically synced ESP32 system time via an Android client, eliminating external RTC chip costs
- Engineered a tuneable control loop allowing the device to dispense user-defined portion sizes at pre-set times, achieving $\pm 1\%$ mass accuracy across 20–100g portions and sub-2s timing precision in a real world demo
- Deployed firmware and build instructions to an **open source repository**, making use of DfMA (Design for manufacturing and assembly) to allow the public to easily modify and build to a use case

IMechE Design Challenge – Autonomous Robotic Device

January – March 2025

- Led the team to develop a bespoke electro-mechanical solution under strict competition rules prohibiting microcontrollers, achieving a **3rd place** finish out of 40+ teams
- Designed a discrete bistable latch circuit with an H-bridge to control the motors, replacing mechanical switches with solid-state switching to enable an automatic timer reset
- Formulated a rigorous test strategy and built a test rig to rapidly iterate vehicle upgrades, leading to an **85%** mass reduction and saving **1.2m** in wiring

Skills

- **Programming Languages** – MATLAB, Python, C++
- **Tools / Software** – Simulink, Fusion 360, Git, Azure DevOps, EasyEDA
- **Manufacturing** – 3D Printing, Laser Cutting, Standard workshop equipment, Soldering